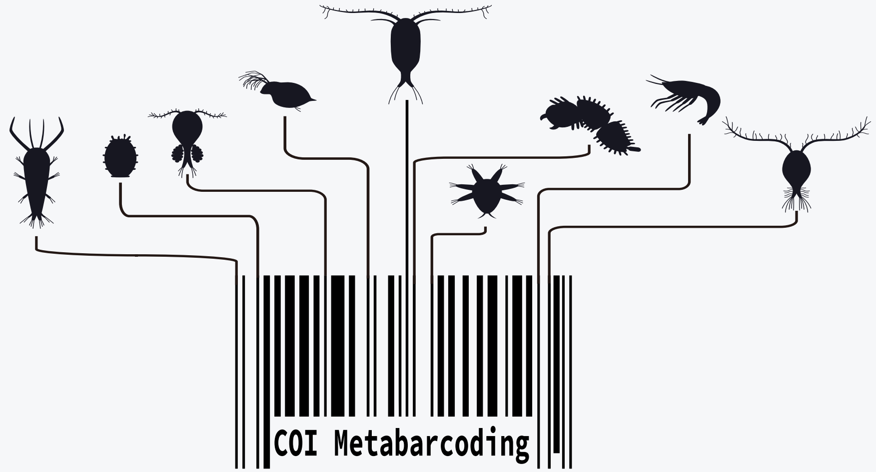
Day 2 · Taxonomy and Community Ecology

Marine eDNA Metabarcoding Analysis using R

Jeju ARMS Case Study

2026.09.01



Song, Chi-une

Postdoctoral Researcher
Department of Life Science
Chung-Ang University

WORKSHOP OVERVIEW

Overview — workshop workflow

Day 1 generated ASVs; Day 2 connects them to ecology.

01
Sampling & DNA Preparation

Collect and prepare the biological material

1. Marine sample collection
2. DNA extraction
3. COI barcode amplification

02
NGS Sequencing

Convert amplified DNA into sequencing reads

1. Library preparation
2. Illumina sequencing
3. Raw sequencing output

03
Bioinformatics Processing

Turn raw reads into reliable sequence variants

1. FASTQ quality control
2. Filtering & denoising
3. ASV inference with DADA2

04
Ecological Analysis

Connect sequence variants to community ecology

1. Taxonomic assignment
2. Alpha & beta diversity
3. Ecological interpretation

DAY 1 · Raw reads → ASVs

DAY 2 · ASVs → Ecology

Today's four questions

ASV sequences are the starting point, not the final answer.

Taxonomy & filtering → α-diversity → β-diversity → Indicator taxa

1. What organisms match our ASVs, and which should we analyze?

→ BLAST search + Taxonomic & ecological filtering

2. How diverse is each community?

→ Rarefaction + α-diversity

Richness · Shannon · Simpson

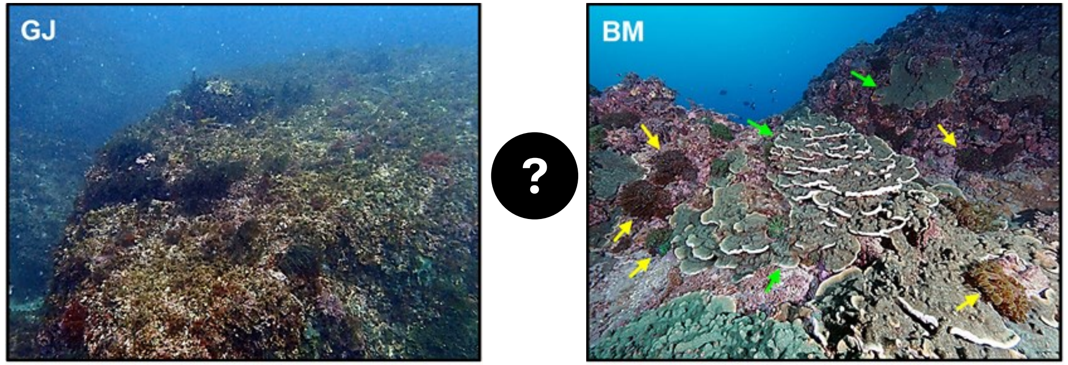
3. How do the two communities differ?

→ β-diversity

Bray–Curtis · nMDS · ANOSIM

4. Which taxa characterize each site?

→ Indicator Value (IndVal)



Gangjeong vs Bomok

KEY POINT · **Each analysis answers a different biological question.**

Two Jeju habitats

Gangjeong and Bomok represent contrasting benthic settings.

GJ

Gangjeong

Macroalgae-dominated habitat

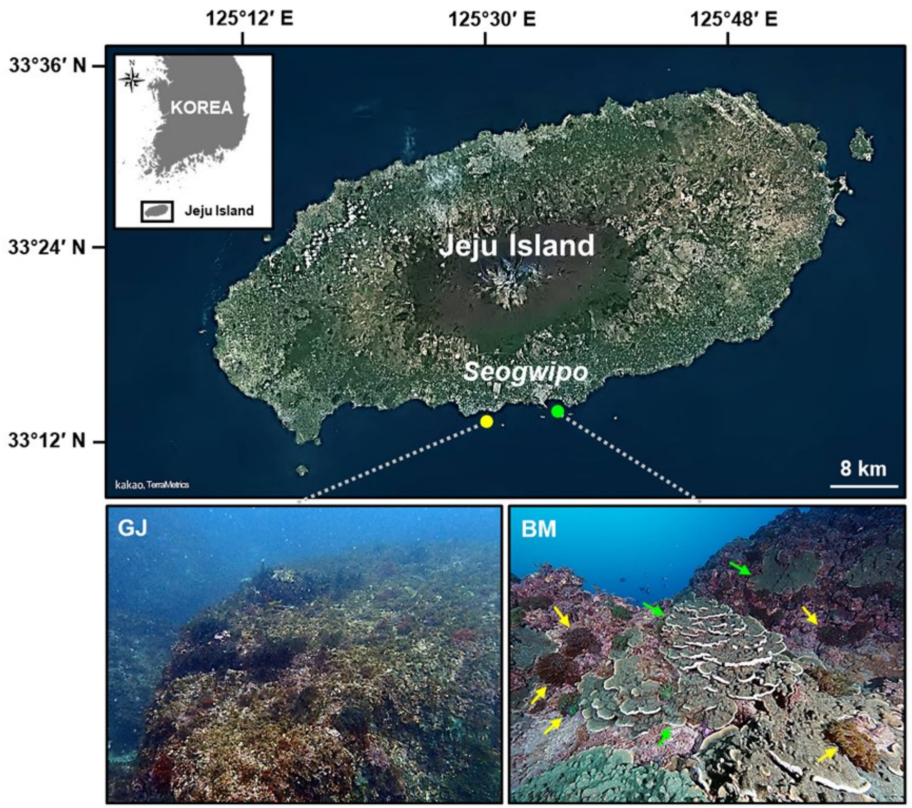
- Turf-forming algae
- Gangjeong1–Gangjeong4

BM

Bomok

Coral-dominated habitat

- Scleractinian corals
- Bomok1–Bomok4



Study sites on the southern coast of Jeju Island

Lee et al. (2024)

KEY POINT · **Eight teaching profiles are compared as two groups of four.**

What Day 1 produced

Day 1 produced an ASV abundance table and representative sequences.

ASV TABLE

- Rows: unique ASVs
- Columns: samples
- Cells: read counts

ASV_ID	GJ1	GJ2	GJ3	GJ4	BM1	BM2	BM3	BM4
ASV25	328	317	228	200	34	24	26	36
ASV26	371	888	356	384	191	57	36	89
ASV28	0	1010	136	101	2734	12201	263	3129
ASV29	420	361	206	416	0	25	25	44
ASV30	91	24	40	22	0	29	10	17
ASV31	44	159	227	270	804	636	1873	883
…								

REPRESENTATIVE SEQUENCES

- One DNA sequence per ASV
- No organism name yet

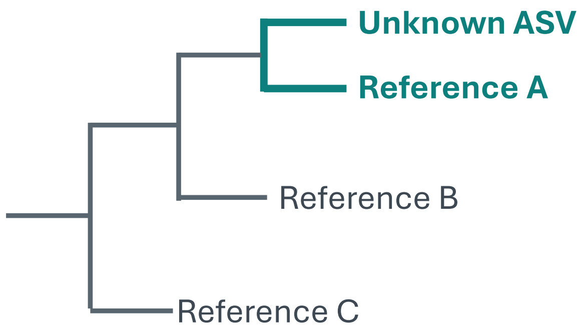
```
>ASV31
AACTTTATATTTAATTTTTGGGGCCTGATCAGCTATAGTAGGAACAGCCTTAAGAATGTT
AATTCGAGCTGAATTAGGTCAACCAGGAAGTTTAATTGGTGATGATCAAATTTATAATGT
TATTGTTACAGCTCACGCTTTTATTATAATTTTCTTTATAGTAATACCTATTATAATTGG
GGGATTTGGAAATTGACTTCTTCCTTTAATACTTGGTGCCCCTGATATGGCATTCCCTCG
ATTAAATAATATGAGATTTTGACTTCTTCCCCCAGC
>ASV452
AACCTTATATTTAATTTTTGGCGCGTTTTCTGGAGTTTTAGGTGCGTGTATTTCTATGTT
AATTCGTATGGAATTAGCTCAACCAGGCAATCACTTACTACTTGGTAATCATCAAGTTTA
CAATGTACTTATAACAGCTCATGCTTTTTTAATGATATTTTTTTTAGTTATGCCAGTAAT
GATAGGAGGTTTTGGTAATTGATTAGTTCCTATTATGATTGGTAGCCCTGATATGGCTTT
TCCTCGTTTAAATAATATTAGTTTTTGATTATTACC
…
```

KEY POINT · **An ASV is a sequence variant, not automatically a species.**

Related organisms have similar barcode sequences

Closely related organisms generally have more similar sequences in the same barcode region.

Evolutionary relationship



Closer relationship

→ fewer sequence differences

Barcode comparisons

Unknown ASV	A T G G C A G G T A C C T G A T C A G T	query
Reference A	A T G G C A G G T A C C T G A T C A G C	high similarity
Reference B	A T G A C A G G T T C C T G A T C A G C	lower similarity
Reference C	A C G A T C G A T C C G T A A T C G A C	low similarity

More similar barcode sequence → stronger evidence for taxonomic similarity

KEY POINT · **Sequence similarity provides the basis for taxonomic assignment.**

Reference databases

Taxonomic assignment compares an unknown ASV with identified reference sequences.

QUERY

Unknown ASV (ASV452)

```
>ASV452
AACCTTATATTTAATTTTTGGCGC
GTTTTCTGGAGTTTTAGGTGCGTG
TATTTCTATGTTAATTCGTATGGA
ATTAGCTCAACCAGGCAATCACTT
ACTACTTGGTAATCATCAAGTTTA
CAATGTACTTATAACAGCTCATGC
TTTTTTAATGATATTTTTTTTAGT
TATGCCAGTAATGATAGGAGGTTT
TGGTAATTGATTAGTTCCTATTAT
GATTGGTAGCCCTGATATGGCTTT
TCCTCGTTTAAATAATATTAGTTT
TTGATTATTACC
```

Compare

REFERENCE DATABASE

Mesophyllum sp. 4 CP-2018 voucher VPF00480 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial

GenBank: MF133365.1

FASTA Graphics

Go to:

```
LOCUS       MF133365                 664 bp    DNA     linear   PLN 21-APR-2018
DEFINITION  Mesophyllum sp. 4 CP-2018 voucher VPF00480 cytochrome oxidase
            subunit 1 (COI) gene, partial cds; mitochondrial.
ACCESSION   MF133365
VERSION     MF133365.1
KEYWORDS    .
SOURCE      mitochondrion Mesophyllum sp. 4 CP-2018
  ORGANISM  Mesophyllum sp. 4 CP-2018
            Eukaryota; Rhodophyta; Florideophyceae; Corallinophycidae;
            Hapalidiales; Hapalidiaceae; Mesophylloideae; Mesophyllum.
        1 aaccttatat ttaatttttg gcgcgttttc tggagtttta ggtgcgtgta tttctatgtt
       61 aattcgtatg gaattagctc aaccaggcaa tcacctacta cttggtaatc atcaagttta
      121 taatgtactt ataacagctc atgctttttt aatgatattt tttttagtta tgccagtaat
      181 gataggaggt tttggtaatt gattagttcc tattatgatt ggtagtcccg atatggcctt
      241 tcctcgttta aataatatta gtttttgatt attaccgcct tcattatgtc ttttattagc
      301 ttcttctatt gttgaagtag gcgcaggtac aggatgaaca gtgtacccac ccttaagttc
      361 tatccaaagt cattctggag catcagtgga cttagcaatt tttagtttac atttagcagg
      421 agcatcttca atacttggtg ctattaattt tatatcaact attttaaata tgcgtaatcc
      481 tggtcaaagt atgtatagaa ttcctttatt tgtttgatca atttttgtta ctgcattttt
      541 acttttattg gcagtacctg ttttagctgg tgcaattaca atgttattaa ctgatagaaa
      601 ttttaatact tcgttttttg atcctgcagg aggaggtgat ccaattttgt atcaacattt
      661 attt
```

Taxonomic classification

DNA barcode sequence

National Center for Biotechnology Information
NCBI

Mesophyllum sp. 4 CP-2018 voucher VPF00480 cytochrome oxidase subunit - Nucleotide - NCBI

KEY POINT · **Reference databases link known DNA sequences to taxonomic names.**

What BLAST does

BLAST rapidly searches a large database for sequences similar to our query.

1 WHAT IS BLAST?

- **B**asic **L**ocal **A**lignment **S**earch **T**ool
- Finds locally similar sequence regions in a large database.

2 HOW DOES IT WORK?

Query sequence → find short matching regions → extend promising matches → rank the strongest hits.

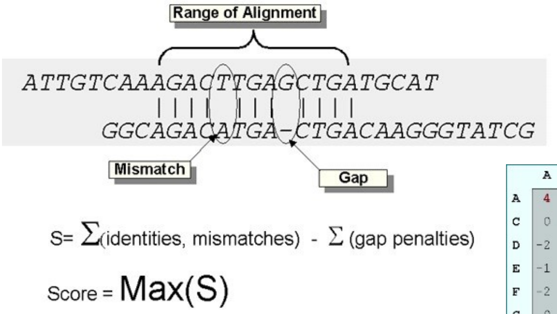
3 WHY IS IT USEFUL?

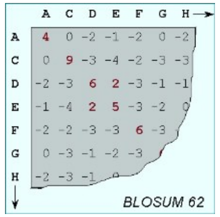
- Fast database search
- Finds similar regions without requiring full-length alignment
- Reports quantitative match metrics

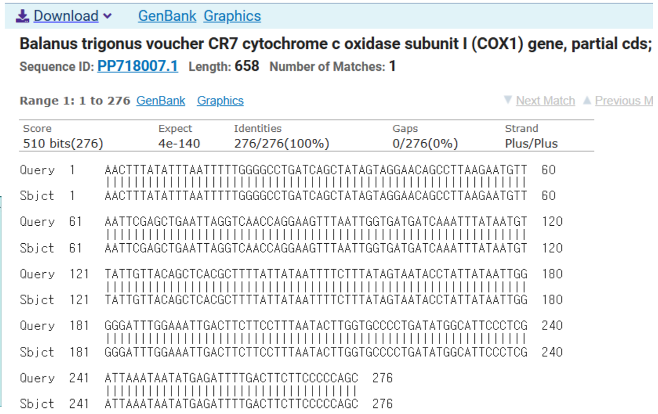
SEARCH PRINCIPLE

BLAST is a local alignment search

- It first finds promising short sequence matches and then extends them to identify stronger local alignments.







KEY POINT · **BLAST rapidly finds and ranks similar reference sequences for an unknown query.**

Reading a BLAST result

Use several metrics to evaluate whether a match is informative.

Two values determine acceptance in this workshop dataset: **percent identity** and **query coverage**.

Key BLAST metrics

Percent identity – matching base within the aligned region

Query coverage – how much of the ASV sequence is aligned

Alignment length – number of bases compared

E-value – expected matches occurring by chance

Bit score – Overall strength of the alignment

Key BLAST metrics

Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenBank Graphics Distance tree of results MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Balanus trigonus voucher CR7 cytochrome c oxidase subunit I (COX1) gene, partial cds; mitochond...	Balanus trigonus	510	510	100%	4e-140	100.00%	658	PP718007.1
Balanus trigonus voucher cr10 cytochrome c oxidase subunit I (COX1) gene, partial cds; mitochondrial	Balanus trigonus	510	510	100%	4e-140	100.00%	658	PP718004.1
Balanus trigonus mitochondrion, complete genome	Balanus trigonus	510	510	100%	4e-140	100.00%	15560	MW646099.1
Balanus trigonus voucher UF:Invertebrate Zoology:45507-Arthropoda cytochrome c oxidase subunit...	Balanus trigonus	510	510	100%	4e-140	100.00%	658	MW277822.1
Balanus trigonus voucher CR9 cytochrome c oxidase subunit I (COX1) gene, partial cds; mitochond...	Balanus trigonus	510	510	100%	4e-140	100.00%	658	PP718008.1
Balanus trigonus voucher BHKG-1689 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitoch...	Balanus trigonus	510	510	100%	4e-140	100.00%	658	PP652141.1

coverage identity

KEY POINT · **High identity over a very short alignment can still be misleading.**

Why identity alone is not enough

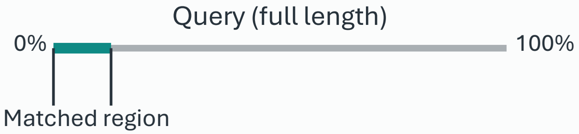
Interpret percent identity together with query coverage.

Two example matches

Example A

100% identity

10% query coverage



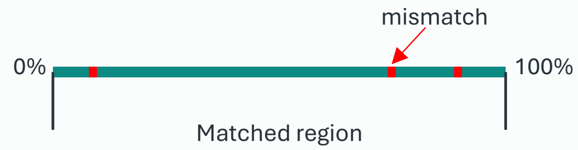
Only a short region matched

→ **Weak evidence**

Example B

97% identity

100% query coverage



Nearly the full ASV matched

→ **Stronger evidence**

- A perfect match over a tiny region can be less informative than a near-perfect match over the full sequence.

The acceptance rule

- Query coverage ≥ **85%**
- Percent identity threshold depends on **taxonomic level**

Species	≥ 98%
Genus	≥ 95%
Family	≥ 90%
Order	≥ 85%

Why does identity start at 85%?

Because this workshop retains assignments down to the **order level**.

⚠ These thresholds are practical criteria for this analysis.
They are not universal cut-offs for all taxa or barcode markers.

KEY POINT · **Reliable assignment requires both sufficient coverage and an appropriate identity threshold.**

What happened to 1,052 ASVs?

Let's inspect the taxonomy table and summarize the BLAST outcomes.

PRACTICAL

Inspect the taxonomy table

Load the file, preview its columns, and summarize BLAST status.

```
# Load the taxonomy table
taxonomy <- read.delim(
  "data/Day2/taxonomy_final_ASV.tsv",
  stringsAsFactors = FALSE)

# Preview the table
head(taxonomy)
names(taxonomy)

# How many ASVs?
nrow(taxonomy)

# Summary of BLAST assignment
table(taxonomy$BLAST_status)
```

BLAST assignment summary

One row per ASV · 1,052 total ASVs

322 **Accepted**
Passed the BLAST criteria

692 **Failed cutoff**
Did not meet identity / coverage criteria

38 **No hit**
No reference match was recovered

1,052 ASVs → BLAST filtering → 322 accepted

KEY POINT · **BLAST filtering reduces 1,052 ASVs to 322 accepted assignments.**

Why BLAST is precomputed

The time is reserved for interpreting results in R.

RUNNING LOCAL BLAST REQUIRES

1. **BLAST+ installation**
   Install and configure the BLAST command-line tools
2. **Reference database**
   Download sequence records for the target marker
3. **Database indexing**
   Format the database before searching
4. **System setup**
   Paths and commands differ across operating systems

IN CLASS → read, inspect, and filter the prepared result

```
taxonomy <- read.delim(
  "data/Day2/taxonomy_final_ASV.tsv",
  stringsAsFactors = FALSE
)

# Preview the table
head(taxonomy)
names(taxonomy)

# How many ASVs?
nrow(taxonomy)

# Summary of BLAST assignment
table(taxonomy$BLAST_status)
```

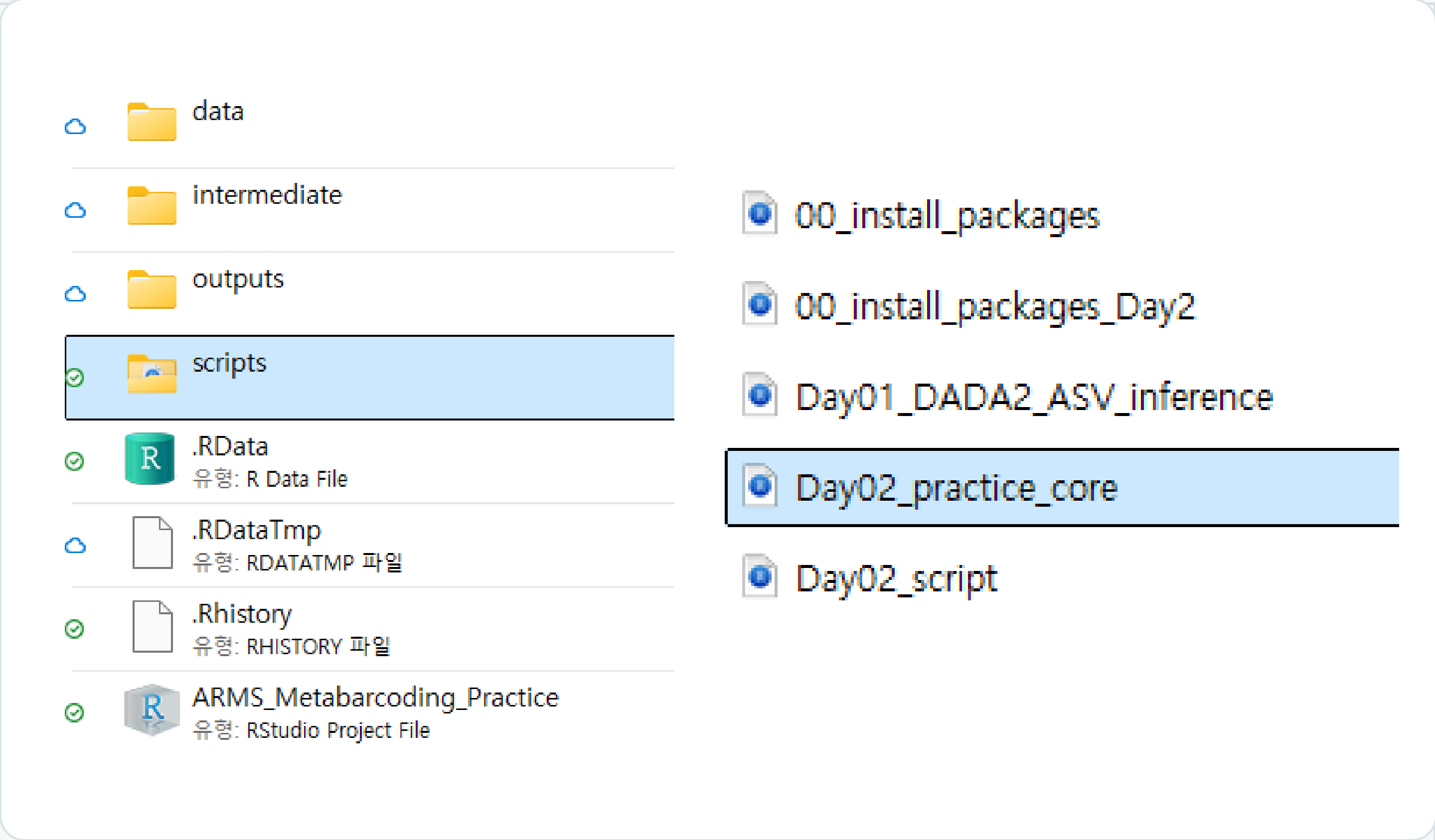
KEY POINT · **No BLAST+ installation or database download is needed in class.**

Open the Day 2 script

PRACTICAL

GOAL

1. Open the Day 2 R project.
2. Load the "Day02_workshop_core.R" script.
3. Confirm the working folder.



GOAL · **Run the project from its root folder and use the short script for live practice.**

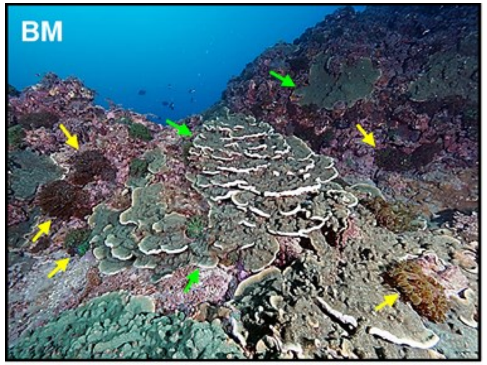
Ecological filtering

A valid BLAST match is not automatically part of the sessile-benthic target community.

KEEP

Sessile-benthic targets

- Accepted ASVs that represent the benthic community targeted in this workshop.
- E.g., Bryozoa, Cnidaria, Porifera, Mollusca, CCA



322 Accepted → 190 Included ASVs

EXCLUDE

- **Contaminants:** Bacteria · fungi · human DNA · other obvious contaminants
- **Non-target organisms:** Fish and other mobile/non-sessile taxa
- **Removed at BLAST filtering:**
  - **No_Hit** — no usable reference match
  - **Failed_cutoff** — insufficient identity and/or coverage

Starting dataset: 1,052 ASVs

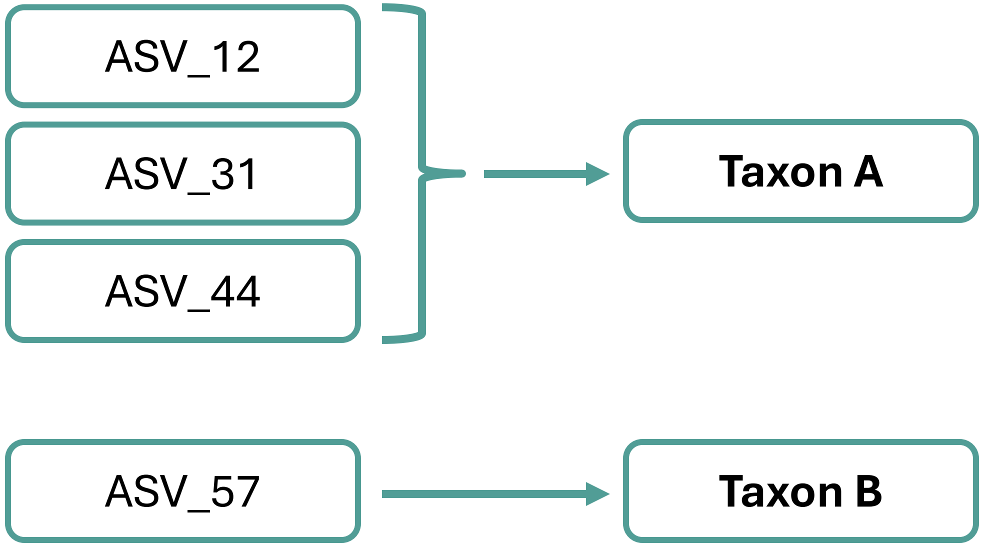
- **730 removed during BLAST filtering**
  → *Failed cutoff or No hit*
- **322 passed BLAST filtering**
  → *132* contaminants */ non-targets*
  → *190 retained for ecological analysis*

KEY POINT · **A valid sequence match and an ecologically relevant target are different decisions.**

ASVs are not taxa

Multiple sequence variants can share the same taxonomic label.

Example of Taxon collapse



Collapse by taxon label

190

included ASVs

↓

Collapse by Taxon_label

↓

88

Individual taxa

KEY POINT · **The ecological matrix uses 88 individual taxa, not 190 ASVs.**

Practical 2 & 3 — Build the community matrix

PRACTICAL

GOAL

- Retain sessile-benthic target ASVs.
- Collapse ASVs sharing the same taxon label.
- Build a "sample × taxon" => community matrix.

Expected size:
8 samples × 88 taxa.

```
# 2. Ecological filtering ---------------------------------
# Select target ASVs
keep <- taxonomy$BLAST_status == "Accepted" &
  taxonomy$Status == "Include"
summary(keep)

# Extract target taxonomy and counts
target_taxonomy <- taxonomy[keep, ]
target_counts <- asv[
  match(target_taxonomy$ASV_ID, asv$ASV_ID),
  sample_names]

nrow(target_taxonomy)    # 190 ASVs

# 3. Collapse ASVs into taxa ------------------------------
taxon_counts <- rowsum(
  as.matrix(target_counts),
  target_taxonomy$Taxon_label)

community <- t(taxon_counts)
dim(community)             # 8 samples x 88 taxa
```

GOAL · **The final community matrix contains 8 samples × 88 individual taxa.**

The community matrix

Rows are samples; columns are the individual ecological taxa.

Community matrix

- Rows are samples
- Columns are the 88 individual taxa
- Values are read counts for each taxon in each sample

	Clytia sp.	*Halecium* sp.	*Laurencia* sp.	*Peyssonnelia* sp.	....
Gangjeong1	1569	4304	0	4805	
Gangjeong2	1552	2398	0	2245	
Bomok1	268	0	10	273	
....					

KEY POINT · **Eight samples × 88 individual taxa; metadata identifies the two sites.**

10-minute break

Keep the Day R project open.

SO FAR

- Taxonomic assignment
- BLAST filtering
- Ecological filtering
- Community matrix building

NEXT

- Rarefaction
- *A*-diversity
- Community composition
- Bray–Curtis and ordination
- Community statistics
- Indicator taxa

KEY POINT · **We have finished data preparation; next we move to community analysis.**

Two kinds of diversity

Within-sample diversity and between-sample differences answer different questions.

α-diversity

How diverse is one sample?

Focus

- Diversity within a community

Examples

- Observed richness
- Shannon
- Simpson

One sample → one diversity value

β-diversity

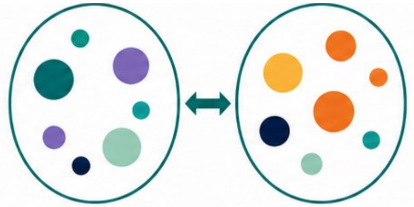
How different are samples?

Focus

- Differences between communities

Our workflow

- Bray-Curtis dissimilarity
- nMDS ordination
- ANOSIM

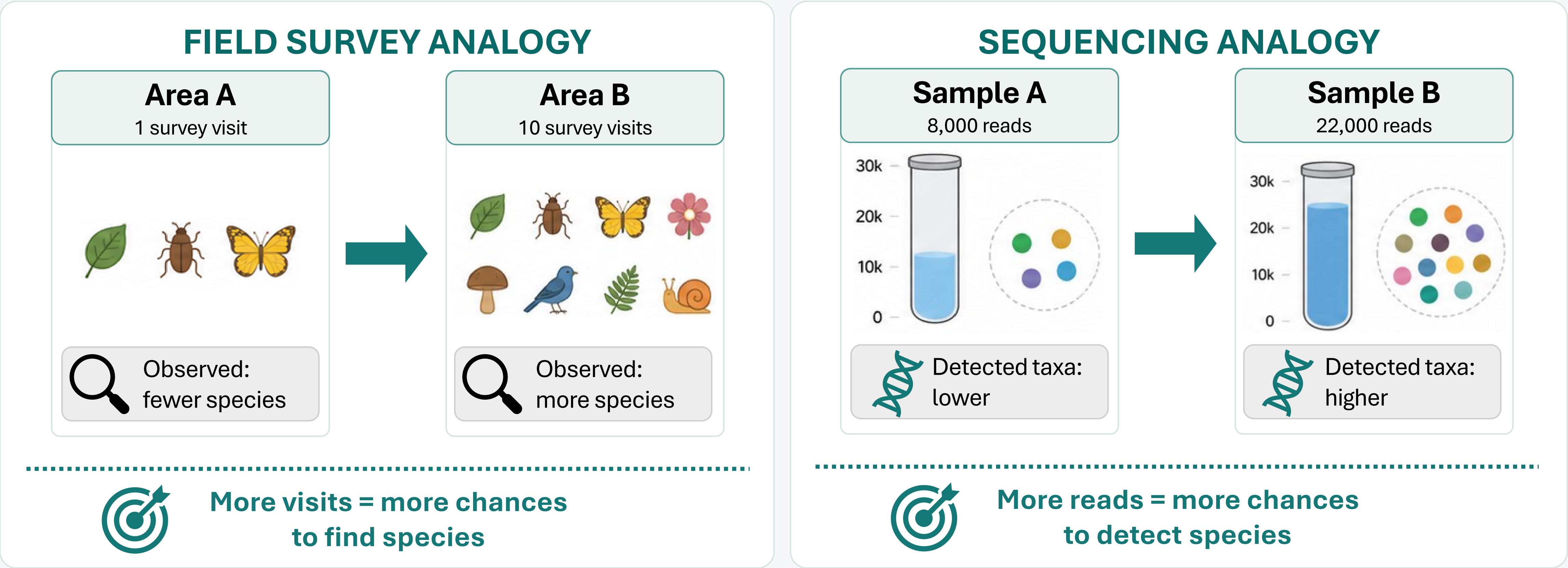


Sample pairs → community differences

KEY POINT · **Alpha describes diversity within a community; beta compares communities.**

Sequencing effort matters

More sampling effort can reveal more taxa even without a true biological difference.



KEY POINT ·

Sequencing effort reflects observation effort; more reads can reveal more taxa even without a true ecological difference.

What rarefaction asks

How rapidly are new taxa recovered as reads accumulate?

TWO QUESTIONS

Q1: Was sequencing depth sufficient?

Curve approaches a plateau

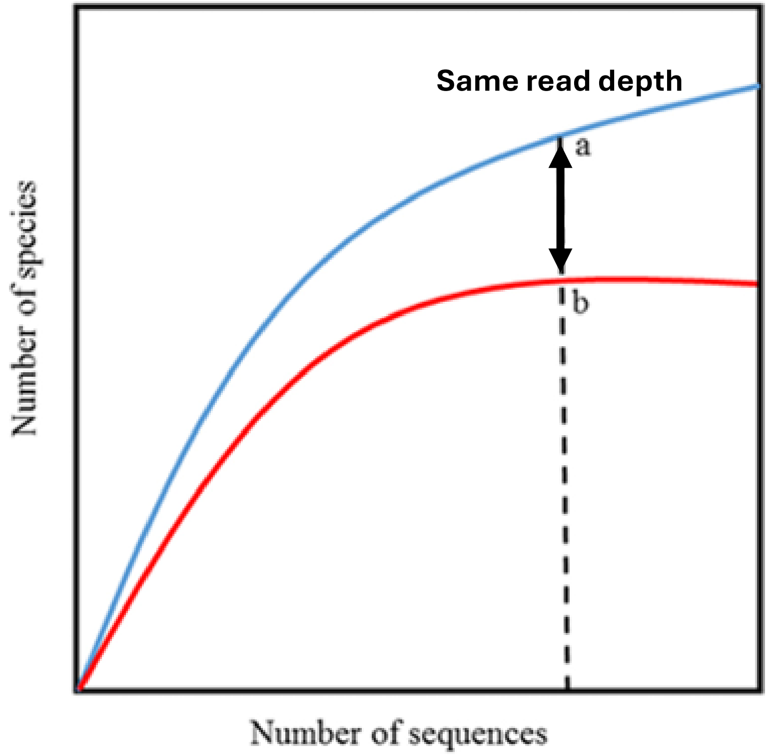
→ Few new taxa are being recovered

Q2: Can samples be compared fairly?

Different read depths provide different opportunities to detect taxa

→ Compare samples at the **same read depth.**

Rarefaction curve



KEY POINT · **Rarefaction helps evaluate sequencing sufficiency and compare richness at a common sampling effort.**

Practical 4 & 5 — rarefaction

PRACTICAL

GOAL

1 · Inspect sequencing depth

Compare read depth among samples.

2 · Check richness recovery

Do the curves begin to flatten?

3 · Standardize sampling effort

Rarefy all samples to 8,499 reads.

WHAT DO YOU NOTICE?

- Read depth differs among samples.
- Most curves begin to flatten.
- The smallest library has **8,499** reads.

```r
# 4. Rarefaction ---------------------------------------------
site_colors <- c(Gangjeong = "#2C7FB8",Bomok = "#E67E22")
rarecurve(community, step = 100, label = FALSE,
          col = site_colors[metadata$Site],
          lwd = 2)
legend("bottomright", names(site_colors),
        col = site_colors, lwd = 2, bty = "n")

# 5. Rarefy to equal sequencing depth ------------------------
sample_depth <- rowSums(community)
min_depth <- min(sample_depth)

sample_depth
min_depth

community_rare <- rrarefy(community, sample = min_depth)
rowSums(community_rare)
```

GOAL · **Standardize sequencing effort before comparing alpha diversity.**

α-diversity measures

Different measures capture different aspects of diversity within a sample.

HOW THE MEASURES DIFFER

Observed richness

- Number of detected taxa
- Ignores relative abundance

Shannon index

- Reflect richness + evenness
- Higher when taxa are more **evenly** distributed

Simpson index (*D*)

- Reflects richness + evenness
- More sensitive to **dominance**
- In vegan, higher value = higher diversity (1-D)

Example: Same richness, different evenness

Measure	Community 1	Community 2	Interpretation
Observed richness	5	5	Same richness
Shannon	1.61	1.18	Higher in Community 1
Simpson (1-D)	0.80	0.60	Higher in Community 1

KEY POINT ·

Practical 6 — α-diversity

PRACTICAL

GOAL

- **Calculate Observed richness, Shannon, and Simpson indices**
- **Compare Gangjeong and Bomok**
- **Run exact Wilcoxon tests**

RUN IN RSTUDIO

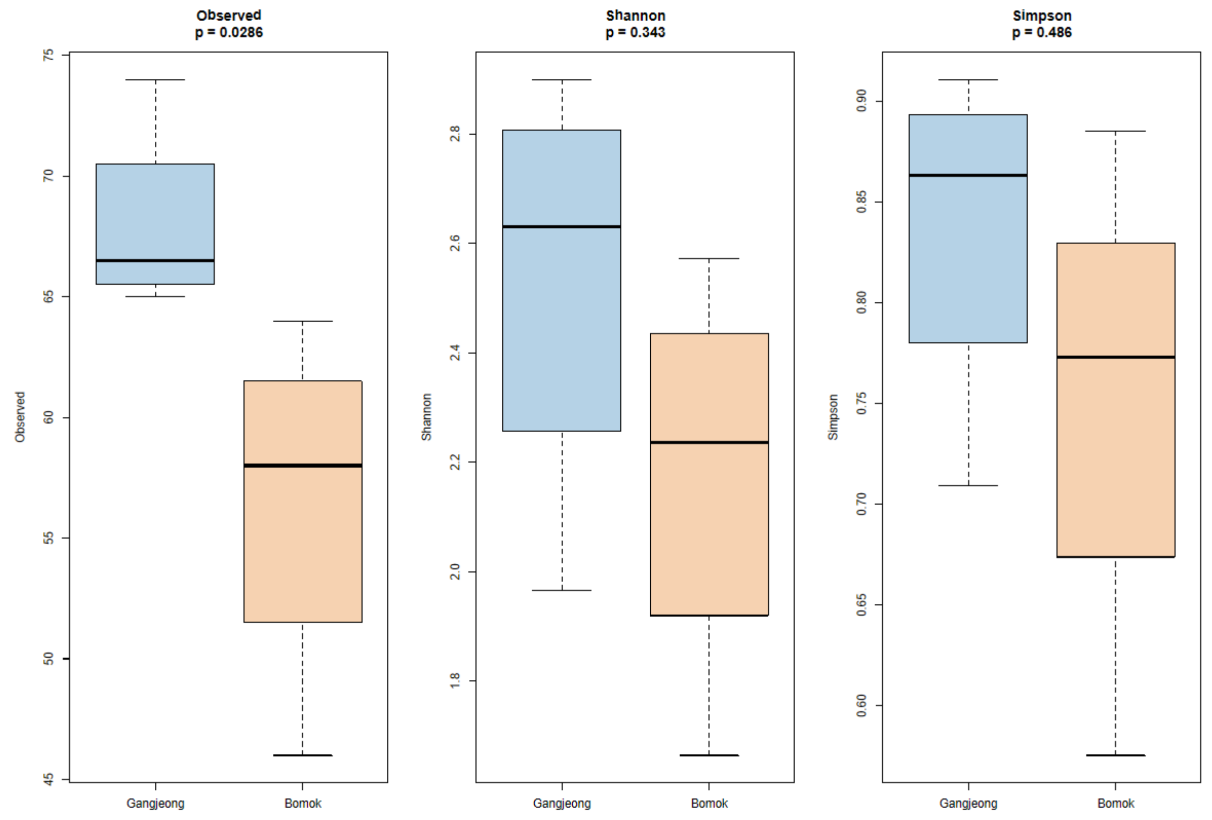
- Execute the code and inspect the table and boxplots.

```
# 6. Alpha diversity ------------------------------------------
alpha <- data.frame(
  SampleID = sample_names,
  Site = metadata$Site,
  Observed = specnumber(community_rare),
  Shannon = diversity(community_rare, "shannon"),
  Simpson = diversity(community_rare, "simpson"))
alpha

par(mfrow = c(1, 3))

for (metric in c("Observed", "Shannon", "Simpson")) {
  test <- wilcox.test(alpha[[metric]] ~ alpha$Site, exact = TRUE)
  boxplot(alpha[[metric]] ~ alpha$Site,
          col = adjustcolor(site_colors, alpha.f = 0.35),
          main = paste0(metric, "\np = ", signif(test$p.value, 3)),
          xlab = "", ylab = metric)
}
par(mfrow = c(1, 1))
```

GOAL · **Generate the alpha-diversity results for all eight samples.**

What do the α-diversity results show?

Only observed richness differed significantly between the two sites.



RESULTS SUMMARY

Observed richness

- GJ > BM (68.0 vs. 56.5)
- *p*-value = 0.0286

Shannon diversity

- GJ ≈ BM (2.532 vs. 2.178)
- ***p-value = 0.343***

Simpson diversity (1-D)

- GJ ≈ BM (0.8366 vs. 0.7516)
- ***p-value = 0.486***

WHAT DOES THIS MEAN?

- Gangjeong contained **more detected taxa**, but overall evenness and dominance structure did not show significant site differences.

KEY POINT · **Gangjeong had higher observed richness, while Shannon and Simpson diversity did not differ significantly.**

Practical 7 — Community composition

PRACTICAL

GOAL

1 · Aggregate target ASVs by Group

Preserve each ASV's broad-group assignment.

2 · Convert counts to proportions

Calculate relative read proportion within each sample.

3 · Visualize community composition

Draw a stacked bar plot for all eight samples.

WHAT SHOULD WE LOOK FOR?

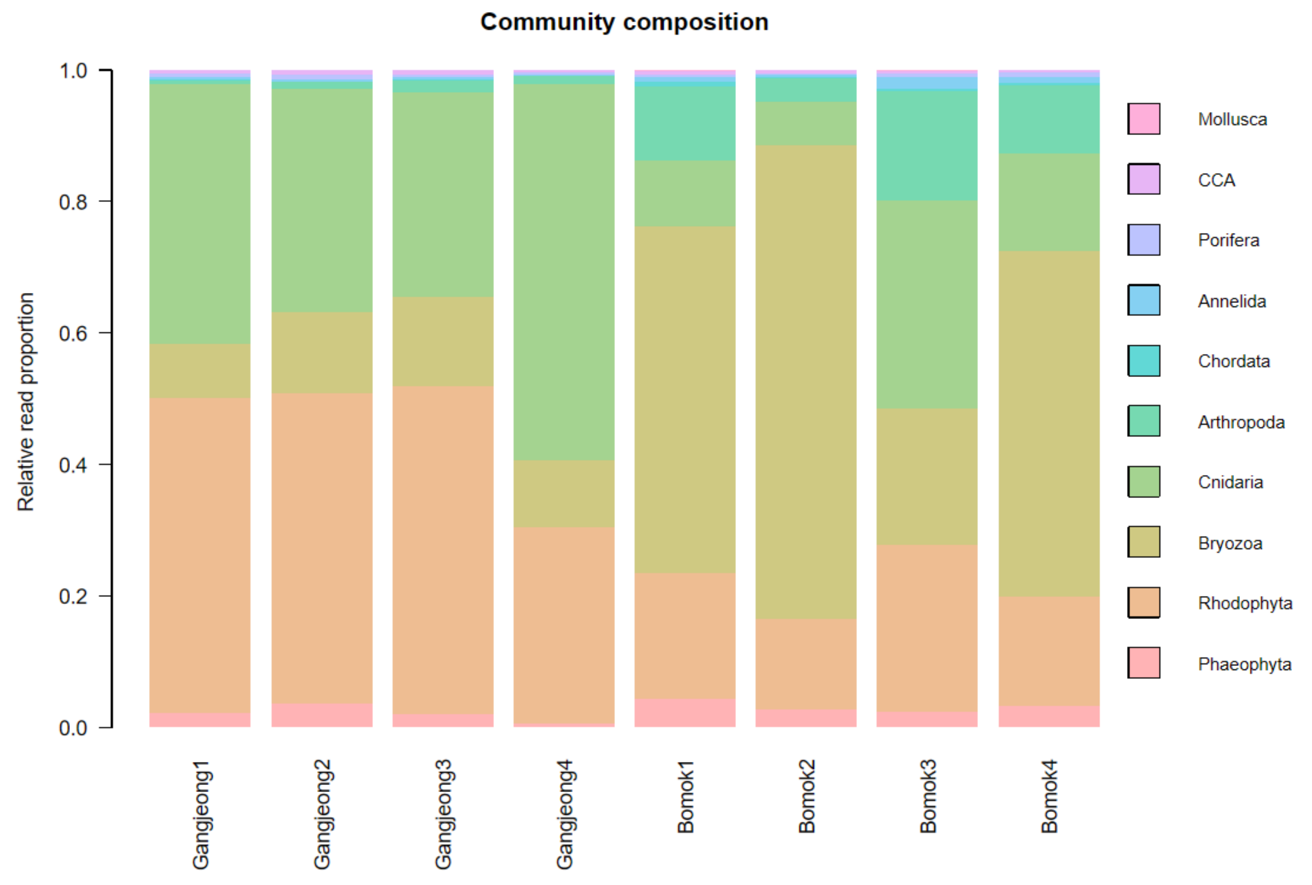
- Which groups dominate each sample?
- Do the two sites differ visually?
- How much variation occurs within each site?

```
# 7. Community composition ------------------------------------
composition_counts <- rowsum(
  as.matrix(target_counts),
  target_taxonomy$Group,
  reorder = FALSE)

composition_relative <- sweep(
  composition_counts, 2,
  colSums(composition_counts), "/")

# Plotting
composition_colors <- hcl.colors(
  nrow(composition_relative), "Set 3")
par(mar = c(9, 5, 4, 10), xpd = TRUE)
barplot(composition_relative,
        col = composition_colors, border = NA, las = 2,
        ylab = "Relative read proportion",
        main = "Community composition")
legend("topright", inset = c(-0.28, 0),
       legend = rev(rownames(composition_relative)),
       fill = rev(composition_colors), bty = "n", cex = 0.85)
```

GOAL · **Visualize which broad biological groups make up each sample.**

What do the composition results show?

Broad-group composition differed visually between the two sites.



RESULTS SUMMARY

Gangjeong

- Mainly composed of **Rhodophyta** and **Cnidaria**
- **Gangjeong4** shows especially high **Cnidaria**

Bomok

- **Bryozoa** dominates several samples.
- Bomok3 is more mixed.

Across samples

- Composition varies within each site.
- The broad pattern still suggests **site-level differences.**

WHAT DOSE THIS MEAN?

- Gangjeong: mainly Rhodophyta and Cnidaria.
- Bomok: often a higher proportion of Bryozoa.
- Samples within a site are not identical.

KEY POINT · **Broad composition suggests site-level differences, but sample-to-sample variation is also evident.**

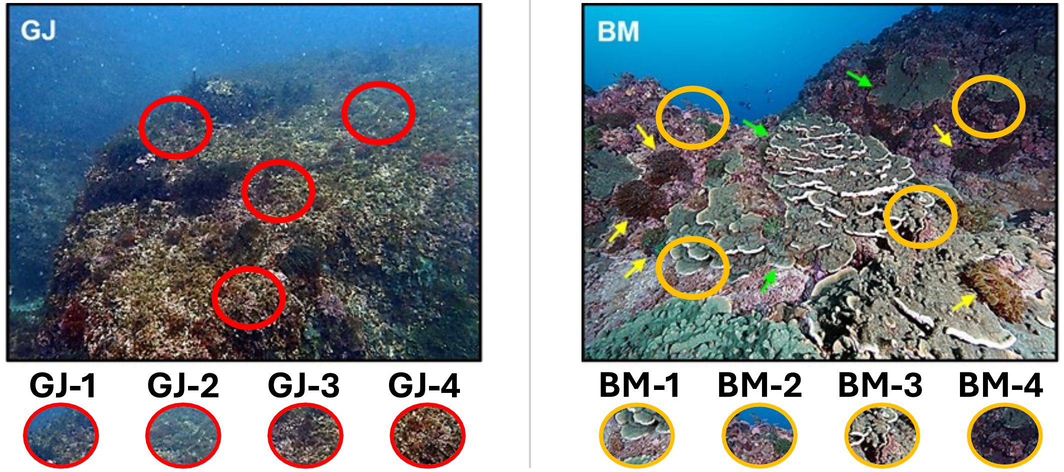
What β-diversity asks?

Do samples contain similar taxa in similar proportions?

WITHIN EACH SITE

Are samples from the same site similar?

- Do they contain similar taxa?
- Are those taxa present in similar proportions?

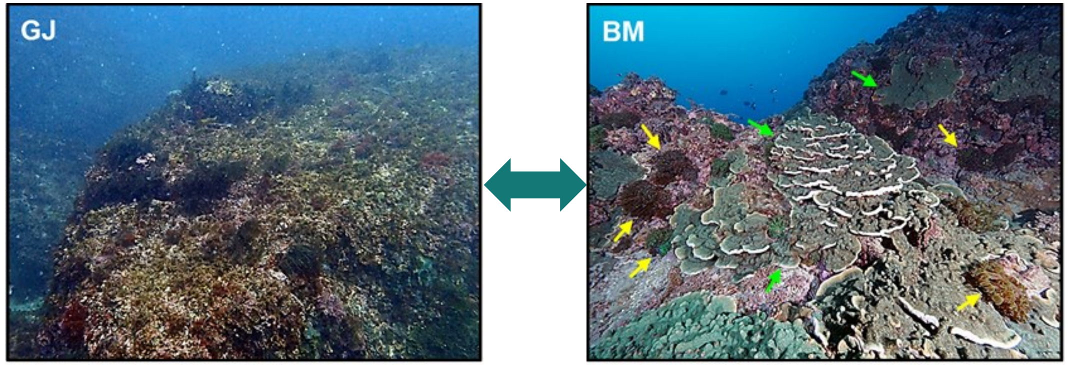


Within-site similarity

BETWEEN SITES

Are Gangjeong and Bomok different?

- Do they contain different taxa?
- Do the relative abundances of shared taxa differ?



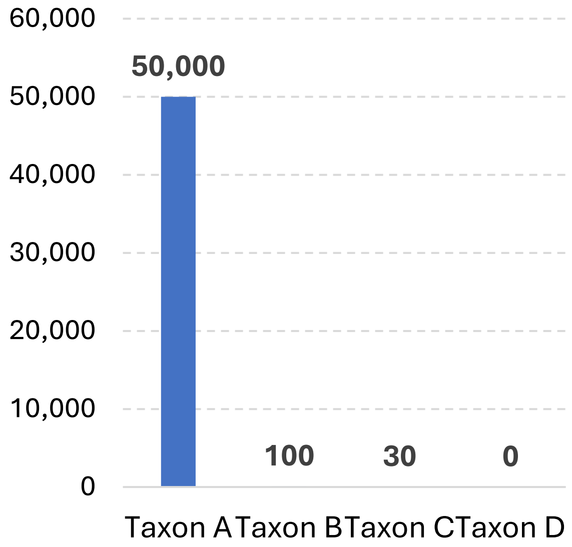
Between-site difference

KEY POINT · **β-diversity compares differences in taxon membership and community composition among samples.**

Why use log1p?

Large read counts can dominate abundance-based distances.

RAW COUNTS



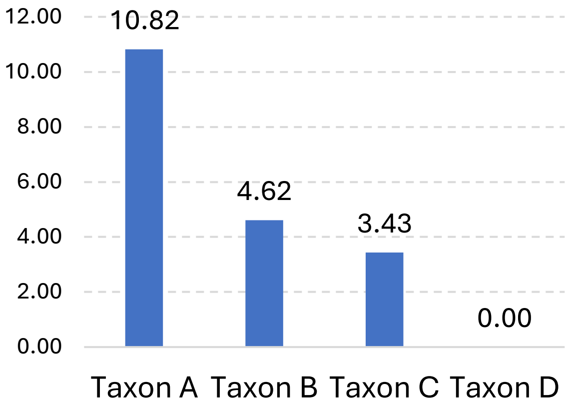
Raw read counts

Taxon A	50,000
Taxon B	100
Taxon C	30
Taxon D	0

- Taxon/sample read counts can differ by orders of magnitude
- A few very abundant taxa may dominate Bray-Curtis distances

AFTER log1p TRANSFORMATION

$$\log 1p(x) = \log_e(x+1)$$



Taxon	Raw counts	log1p(x)
A	50,000	**10.82**
B	100	**4.62**
C	30	**3.43**
D	0	**0.00**

- Compress large count differences
- Reduces dominance of extremely abundant taxa

Why '+1'?

- $log(0)$ is undefined
- Zero counts remain zero
- 0 → $log(1)$ = 0

KEY POINT · **log1p() reduces extreme count differences while preserving zeros before Bray–Curtis analysis.**

What is Bray–Curtis dissimilarity?

A single number summarizes how different two communities are.

PAIRWISE COMPARISON (Example)

Sample A (Site 1)

Taxa	Abundance
Taxon 1	40
Taxon 2	30
Taxon 3	10

Total abundance = 80

Compare the abundance of all taxa

Sample B (Site 2)

Taxa	Abundance
Taxon 1	35
Taxon 2	5
Taxon 3	25

Total abundance = 65

HOW BRAY-CURTIS IS CALCULATED

Taxon	A	B	min(A, B)
Taxon 1	40	35	35
Taxon 2	30	5	5
Taxon 3	10	25	10
Total	80	65	$C_{AB} = \Sigma min(A,B) = 50$

Bray-Curtis formula

$$BC_{AB} = 1 - \frac{2C_{AB}}{S_A + S_B}$$

Plug in the values

$$BC_{AB} = 1 - \frac{2(50)}{80 + 65} = 1 - \frac{100}{145} = 0.31$$

BRAY–CURTIS DISSIMILARITY

Often referred to as a community "distance"

0 Similar ⟷ **1** different

- Use abundance information
- Calculated for every pair of samples
- Produces a dissimilarity matrix

Interpretation

- 0 → identical composition
- 0 < BC < 1 → more different
- 1 → completely different

Distance (dissimilarity) matrix

	A	B	C
A	0	0.42	0.67
B	0.42	0	0.35
C	0.67	0.35	0

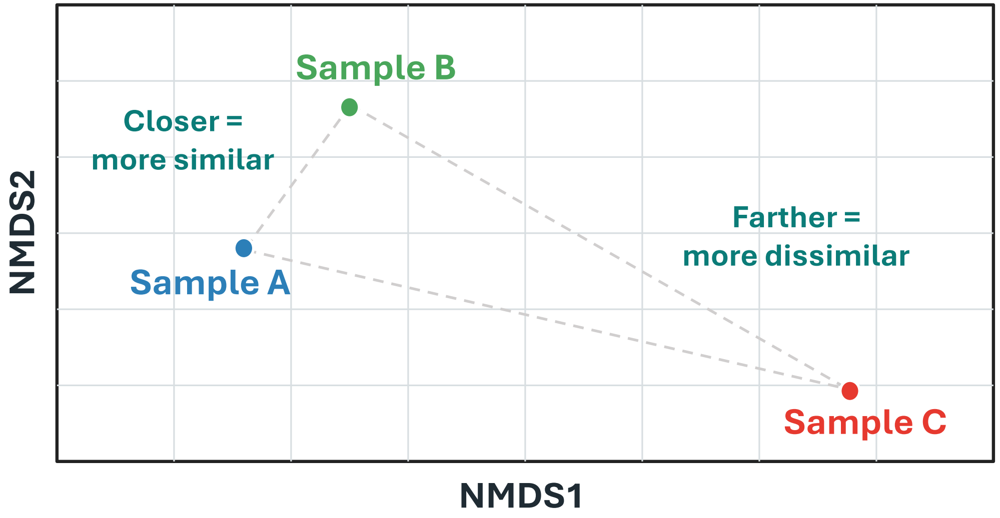
KEY POINT · **Bray–Curtis converts differences in taxon abundance into pairwise community dissimilarities.**

nMDS: Visualizing community dissimilarities

nMDS turns pairwise community dissimilarities into a 2D map while preserving their rank order

WHAT nMDS DOES

Bray–Curtis dissimilarities → **nMDS (2 dimensions)**

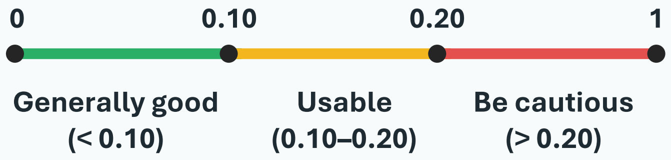


- Each point = one sample (community)
- NMDS1 and NMDS2 have no direct ecological meaning

STRESS: HOW GOOD IS THE FIT?

Stress tells us how well the 2D map represents the original dissimilarity relationships.

Stress value (lower is better)



- Lower stress = better 2D representation
- Always check stress before interpretation
- 2D distances are not exact Bray–Curtis values.

KEY POINT · **Closer points represent more similar communities, and lower stress means a more reliable 2D representation.**

Practical 8 — dissimilarity and nMDS

PRACTICAL

GOAL

1 · Transform the community matrix
Apply log1p() to reduce extreme count differences.

2 · Calculate Bray–Curtis dissimilarity
Compare every pair of samples.

3 · Generate an nMDS ordination
Visualize community relationships in 2D.

4 · Check the stress value
Evaluate how well the ordination represents the dissimilarities.

WHAT SHOULD WE CHECK?

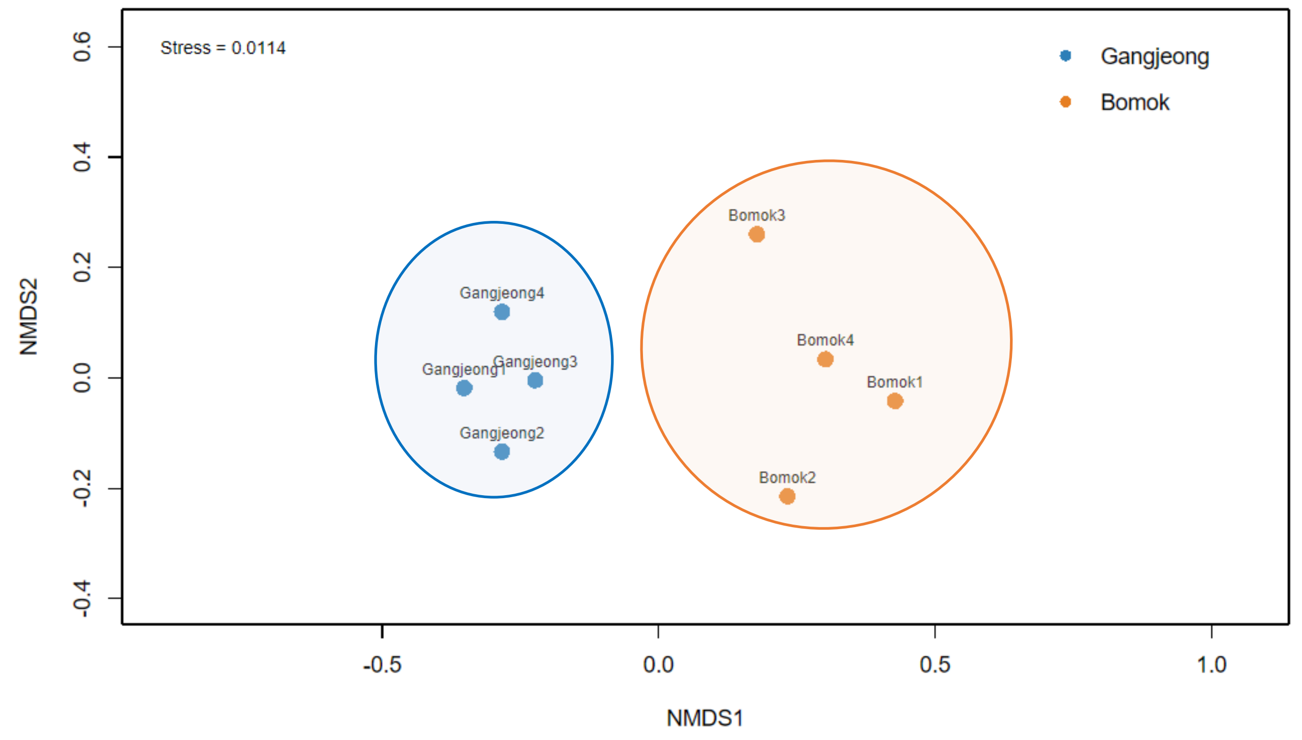
- Do Gangjeong and Bomok separate visually?
- Are samples within each site relatively close?
- Is the stress value sufficiently low?

```
# 8. Bray-Curtis and nMDS ------------------------------------

community_log <- log1p(community)

bray <- vegdist(community_log, method = "bray")
print(bray)

nmds <- metaMDS(community_log, distance = "bray", k = 2, trymax = 100,
                autotransform = FALSE,  trace = FALSE)
nmds$stress

plot(nmds, type = "n")
points(nmds, display = "sites", pch = 19, cex = 1.5,
       col = site_colors[metadata$Site])
text(nmds, display = "sites", labels = sample_names, pos = 3, cex = 0.7)
legend("topright", names(site_colors), col = site_colors, pch = 19, bty = "n")
```

GOAL · **Calculate and visualize community differences among the eight samples.**

What do the nMDS results show?

Gangjeong and Bomok show clear visual separation in community composition.



RESULTS SUMMARY

Gangjeong

- Samples form a relatively tight cluster
- Located together in nMDS space

Bomok

- Samples are more dispersed than Gangjeong
- Clearly separated from Gangjeong along the main ordination pattern

Stress

- Stress = 0.0114
- Very low distortion in the 2D representation

KEY POINT · **Gangjeong and Bomok appear compositionally distinct, but statistical testing is still required.**

ANOSIM (Analysis of Similarity)

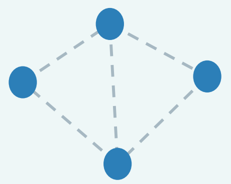
ANOSIM tests whether communities from different groups are more dissimilar than communities within the same group.

1 · IDEA

Compare rank similarities between and within groups.

Within-group

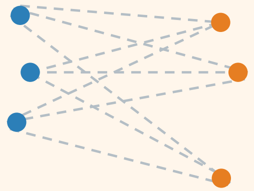
pairs from the same site



Average rank dissimilarity = r_W

Between-group

pairs from different sites



Average rank dissimilarity = r_B

If communities differ by site, between-group similarities (r_B) will be lower than within-group similarities (r_W).

2 · TEST STATISTIC R

ANOSIM uses the ranks of all pairwise dissimilarities.

$$R = \frac{r_B - r_W}{M/2}$$

- r_B :mean rank of between-site dissimilarities
- r_W :mean rank of within-site dissimilarities
- M = total number of pairwise dissimilarities

INTERPRETATION OF R

−1	0	+1
$R < 0$	$R \approx 0$	$R \to 1$
within > between	little separation	strong separation

3 · HOW IT WORKS

1. **Input**
   Bray–Curtis dissimilarity matrix and group labels (e.g., sites).
2. **Rank dissimilarities**
   Order all sample-pair dissimilarities
3. **Compare mean ranks**
   Average ranks for
   Between-site ranks vs. within-site ranks
4. **Compute the R statistic**
   Summarize the separation strength
5. **Assess significance**
   Use permutations of group labels (999) to obtain a *p*-value

- ***R* = separation strength**
- ***p*-value = permutation support**

KEY POINT · **Higher *R* means stronger group separation; the *p*-value tells us whether it is statistically supported.**

Practical 9 — test site effects (ANOSIM)

PRACTICAL

GOAL

1 · Run ANOSIM
Test the difference between Gangjeong and Bomok.

2 · Check the R statistic
Higher R = stronger group separation.

3 · Check the p-value
Does the permutation test support the difference?

WHAT DO YOU NOTICE?

- R = 0.979
- p-value = 0.026

→ Strong separation between the two sites

```
# 9. ANOSIM --------------------------------------------------

anosim_result <- anosim( bray, metadata$Site, permutations = 999)

anosim_result
```

Result

```
> anosim_result

Call:
anosim(x = bray, grouping = metadata$Site, permutations = 999)
Dissimilarity: bray

ANOSIM statistic R: 0.9792  → separation strength
      Significance: 0.026   → statistical support

Permutation: free
Number of permutations: 999
```

KEY POINT · **ANOSIM supports a strong difference in community composition between Gangjeong and Bomok.**

Which taxa characterize a site?

Indicator analysis (IndVal) connects community-level differences to individual taxa.

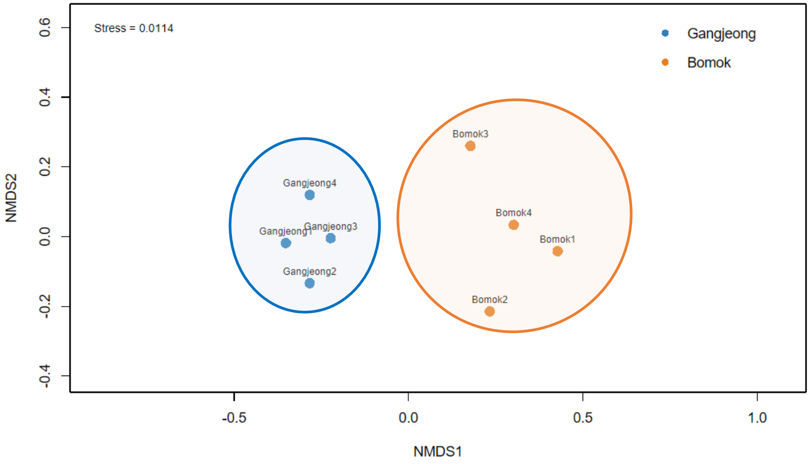
Community-level result

Gangjeong and Bomok differ in overall community composition.

ANOSIM

→ R = 0.979

→ p = 0.026



?

The sites are different
— but which taxa drive this pattern?

Taxon-level question

Which taxa characterize each site?

Indicator analysis (IndVal) looks for taxa that are:

- Strongly associated with one site
- Consistently detected within that site

These two properties are called ***specificity*** and ***fidelity***.

KEY POINT · **Indicator analysis identifies taxa that best characterize each site.**

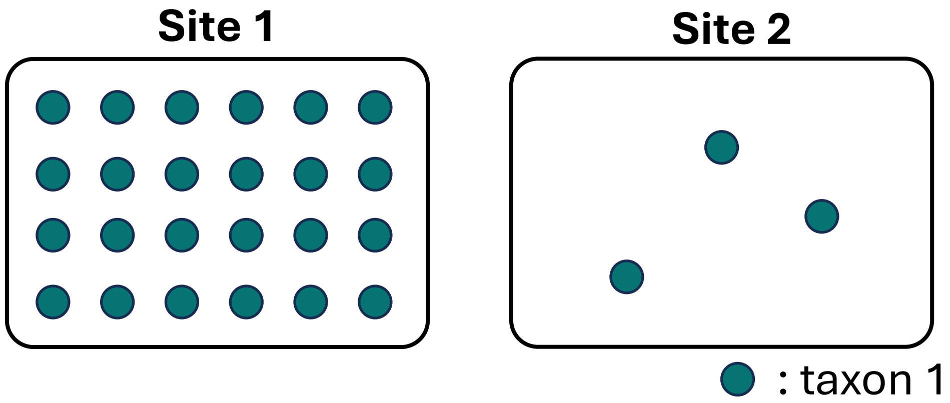
IndVal: Specificity and Fidelity

A useful indicator is both concentrated in one site and consistently present.

$$\text{IndVal} = \sqrt{A \times B}$$

SPECIFICITY (A)

Is this taxon mainly associated with one site?



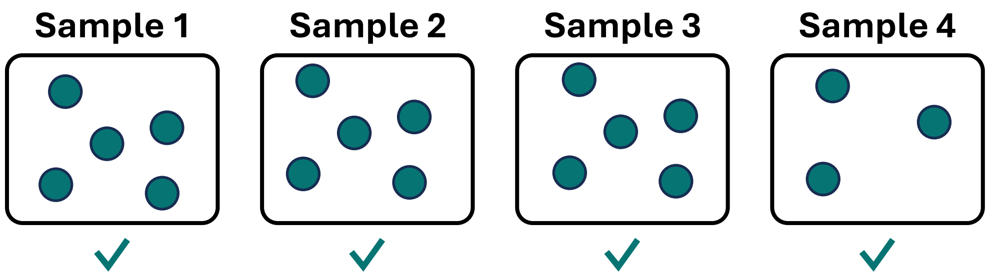
High when most abundance is concentrated in one site.

$$A = \frac{\text{mean abundance in target site}}{\text{sum of mean abundance across all sites}}$$

FIDELITY (B)

Does the taxon occur consistently within that site?

For site 1:



High when the taxon is present in most samples of that site.

$$B = \frac{\text{number of samples where present}}{\text{total samples in the site}}$$

KEY POINT · **A strong indicator taxon has both high specificity and high fidelity.**

Practical 10 — indicator taxa (IndVal)

PRACTICAL

GOAL

1 · Calculate IndVal

Evaluate all taxa for each site.

2 · Inspect A and B

- A = specificity
- B = fidelity

3 · Identify indicator taxa

Look for high IndVal and ***p* < 0.05**.

WHAT SHOULD WE CHECK?

- Which site is the taxon associated with?
- Are A and B both high?
- Is the permutation p-value < 0.05?

```
# 10. Indicator taxa -----------------------------------------
set.seed(20260821)

indval <- multipatt(
  community,
  metadata$Site,
  func = "IndVal.g",
  control = permute::how(nperm = 999))

summary(indval, alpha = 0.05, indvalcomp = TRUE)
```

Specificity

Fidelity

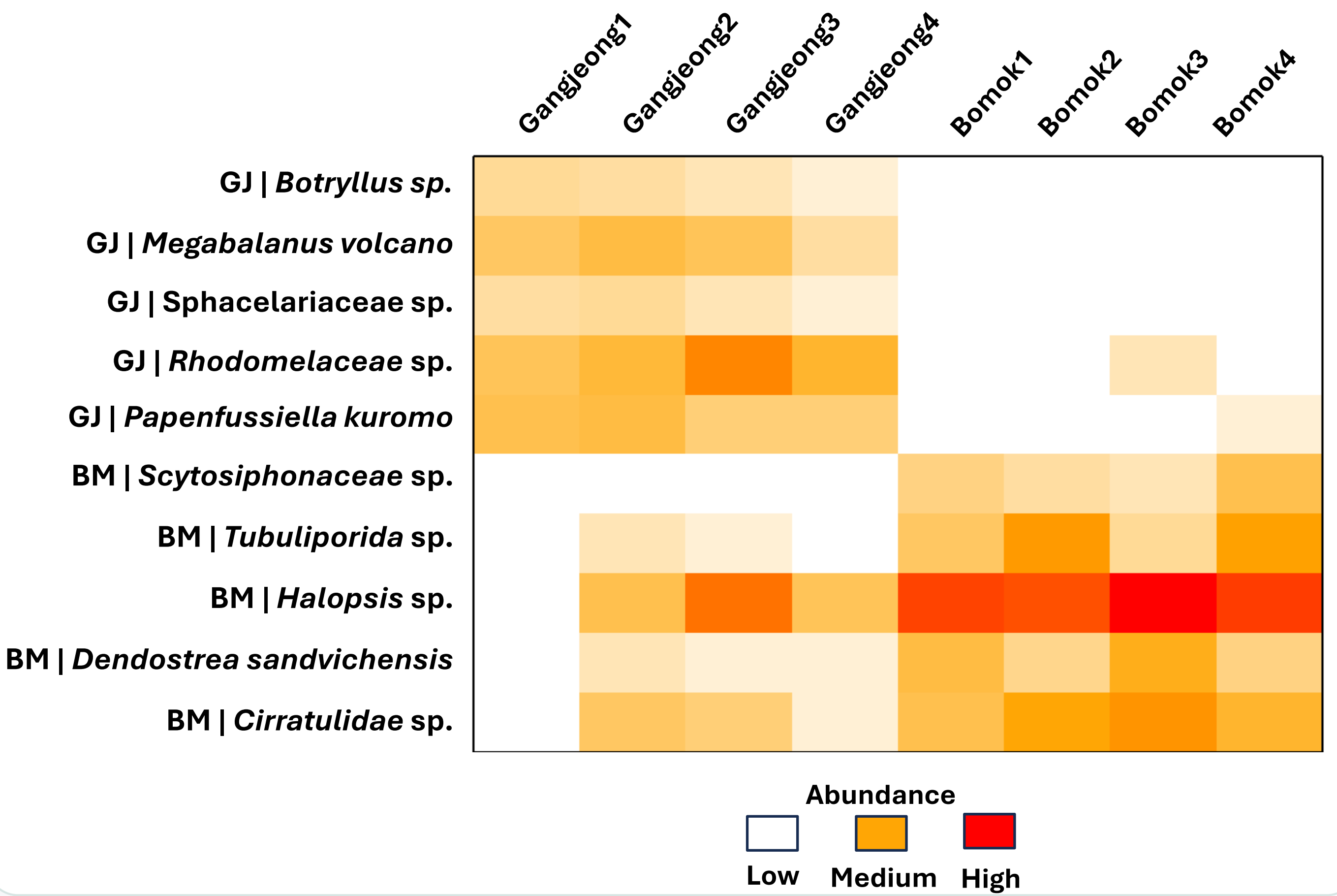
IndVal = $\sqrt{A \times B}$

```
 Group Bomok  #sps.  5
                                A      B  stat p.value
Scytosiphonaceae           1.0000 1.0000 1.000   0.026 *
Tubuliporida               0.9741 1.0000 0.987   0.026 *
Halopsis sp.               0.9529 1.0000 0.976   0.026 *
Dendostrea sandvichensis   0.9310 1.0000 0.965   0.026 *
Cirratulidae               0.8824 1.0000 0.939   0.026 *
```

GOAL · IndVal identifies taxa that are strongly and consistently associated with a site.

What do the indicator-taxon results show?

IndVal links site-level community differences to specific taxa and their broad taxonomic groups.



RESULTS SUMMARY

IndVal pattern

- 20 significant indicators: 15 for Gangjeong and 5 for Bomok
- The **heatmap** shows the **top five taxa** for each site
- Indicators are concentrated in their associated site

Site	Indicator taxon	Group
GJ	*Botryllus sp.*	**Chordata**
	Megabalanus volcano	**Arthropoda**
	Sphacelariaceae	**Phaeophyta**
	Rhodomelaceae	**Rhodophyta**
	Papenfussiella kuromo	**Phaeophyta**
BM	*Scytosiphonaceae*	**Phaeophyta**
	Tubuliporida	**Bryozoa**
	Halopsis sp.	**Cnidaria**
	Dendostrea sandvichensis	**Mollusca**
	Cirratulidae	**Annelida**

KEY POINT · **Indicator taxa identify which taxa best characterize Gangjeong and Bomok.**

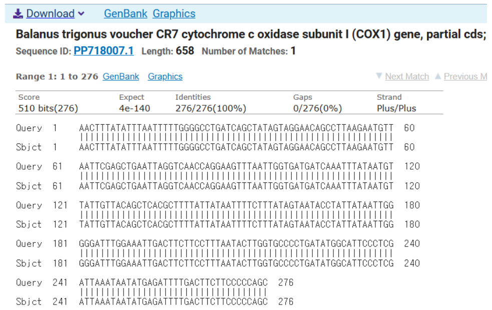
Today’s four questions — answered

From ASVs to ecological interpretation.

1. Taxonomy & filtering

What matched our ASVs?

- 322 passed BLAST filtering
- 190 target ASVs
- 88 collapsed taxa

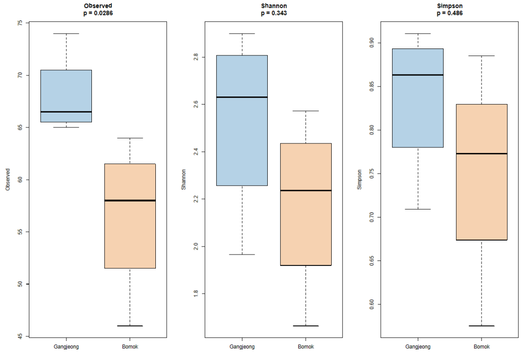


BLAST + ecological filtering + taxonomic collapse

2. *α*-diversity

How diverse was each site?

- Observed:
→ GJ > BM ($p = 0.0286$)
- Shannon and Simpson:
→ not significant.

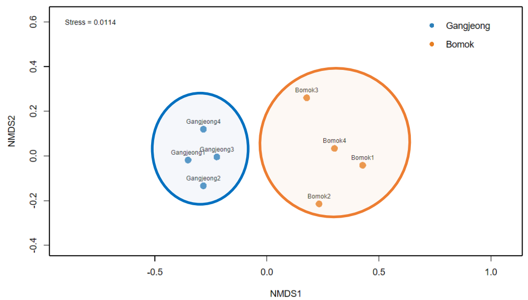


Rarefaction + richness / Shannon / Simpson

3. *β*-diversity

How different were the communities?

- Clear nMDS separation.
- ANOSIM:
$R = 0.979$, $p = 0.026$.
→ Significant site-level difference

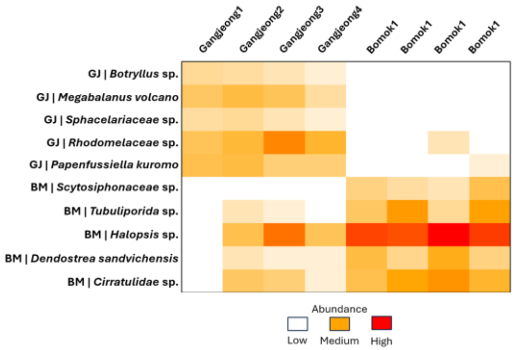


log1p + Bray–Curtis + nMDS + ANOSIM

4. Indicator taxa

Which taxa characterized each site?

- GJ: 15 indicators, mainly algae.
- BM: 5 indicators across benthic groups.



IndVal + indicator-taxon heatmap

KEY POINT · **Each step answers a different question — from taxonomic identity to community ecology.**

- The end -

Thank you!